

POOL VS. DAD ANALYSIS

An Overall Number of Games, Win Rate, Win Condition, Recency, Break Advantage, and Game Location Analysis

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DOCUMENT OVERVIEW AND DISCUSSION OF DATA

The following analysis look at total games of pool played by Ted and his Dad, Tim, since Ted started tracking data. Data collection started on April 10th, 2017. The data includes the following variables: date of game played, location, who breaks, who won, the win condition, and the number of balls remaining for both the winner and the loser.

POOL VS. DAD BREAKDOWN DISCUSSION

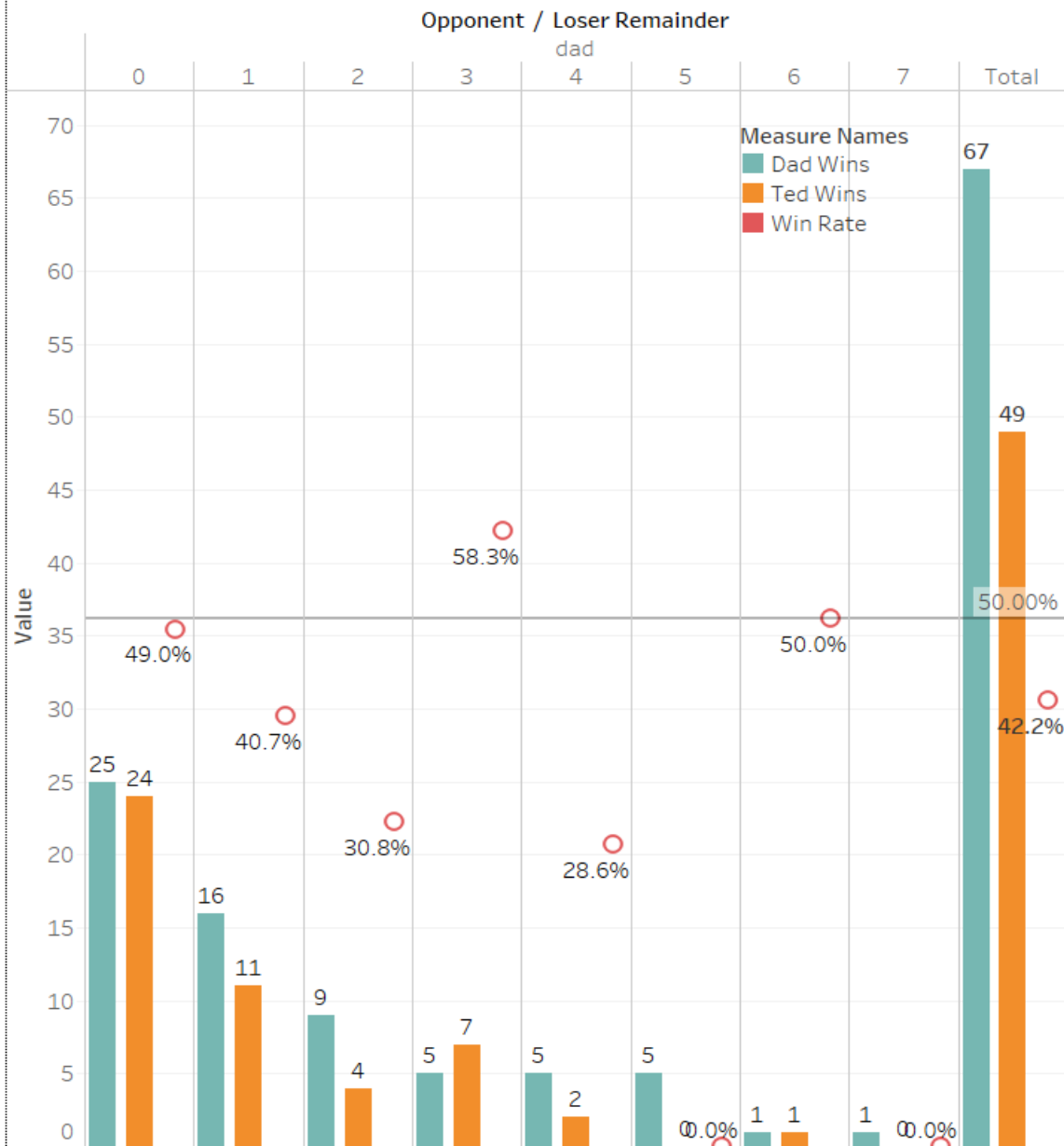
Overall, Ted wins 42.2% of the 116 games played since tracking data. In general, games tend to be close, with fewer observations as the games have a bigger margin of victory. Ted's win-rate is the highest in closer margin games, with Tim taking a bigger share of the games as the margin of victory increases, besides for a 3-ball margin of victory, where Ted has 7 games to Tim's 5 games. Win type is important to consider, as not all games end by the 8-ball being pocketed with no other balls remaining. 75 of 116 games end in regulation, or 64.66%. 15, 22, and 4 games end by scratch on 8, early 8, and wrong pocket, respectively. I would anticipate the share of games ending in regulation to increase as time goes on as both players become more skilled. The recency analysis shows the last 120 days from day of compilation, and it is observed that Ted is winning 62.5% of the games in the last 120 days, or 15 of 24 games. It is also observed that the highest share of games end with 1 ball remaining as opposed to 0 in the overall dataset. I would hypothesize that as the players become more skilled, they are more capable of stringing balls together, perhaps winning the game in a string of made balls causing the opponent to be more likely to have more balls remaining.

IS BREAKING AN ADVANTAGE DISCUSSION

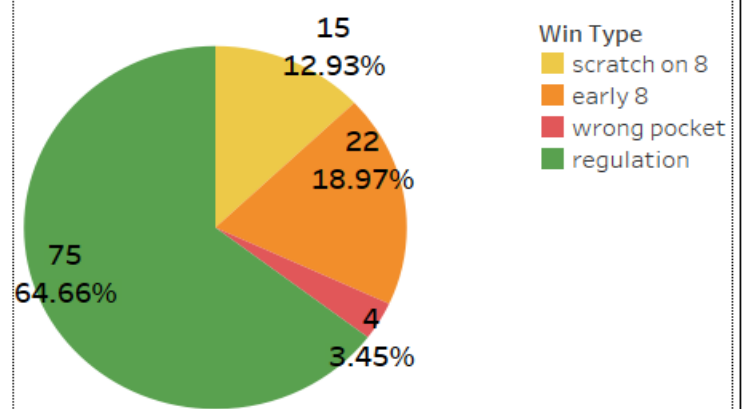
In this image, it is observed that Ted's win-rate is 34.4% when Tim breaks, and 50.9% when he breaks, implying breaking is an advantage. The data for when Ted breaks also seems to be more spread more evenly across different win margins. I believe this to be due to Ted breaking the rack with more power, spreading the balls out more across the table and allowing the playing to string balls together more easily thus leading to victories by larger margins for either player. Opposite of this, when Tim breaks, the win margin is overwhelmingly 0 or 1 balls remaining for the loser, implying games are closer when Tim breaks. I hypothesis this is because of the opposite of when Ted breaks; Tim breaks with less power and the balls are more grouped, causing the players to play a slower game due to difficulty stringing balls together. This leads to games that end with a closer margin of victory than games where Ted breaks.

Pool vs. Dad Breakdown

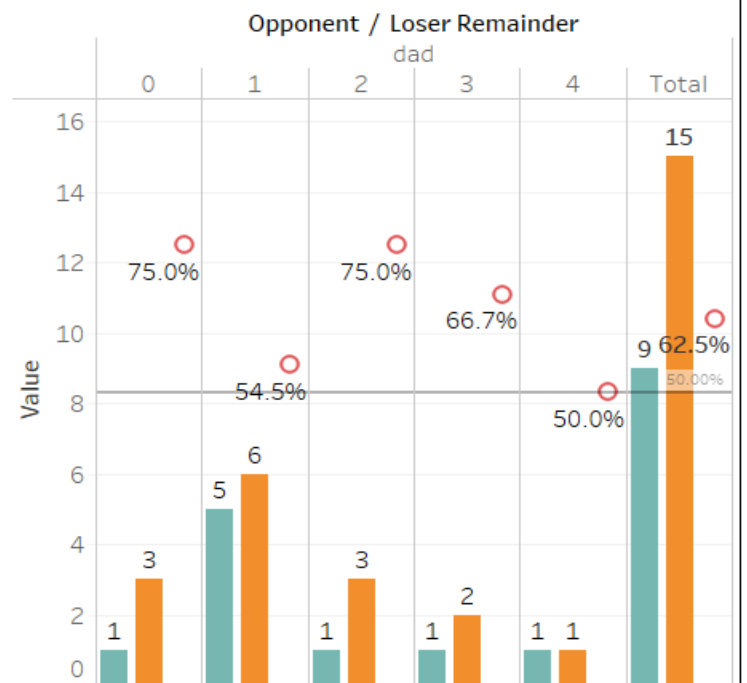
Games&Win Rate at Different Loser Balls Remaining



Win Type vs. Dad

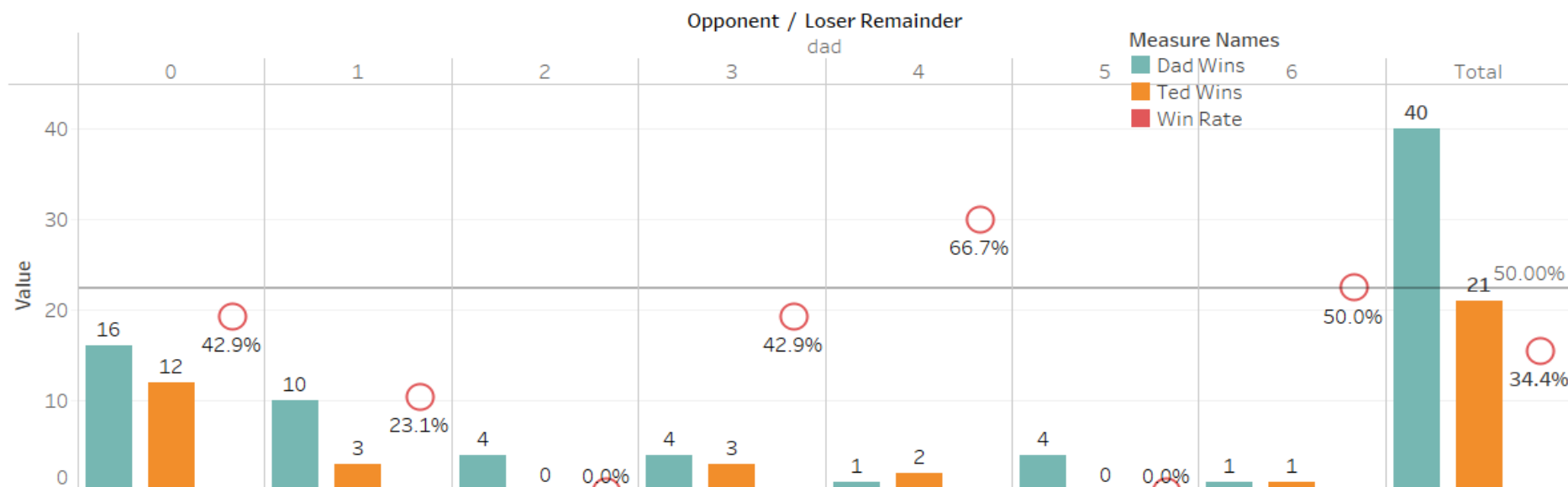


Last 120 Days of Pool

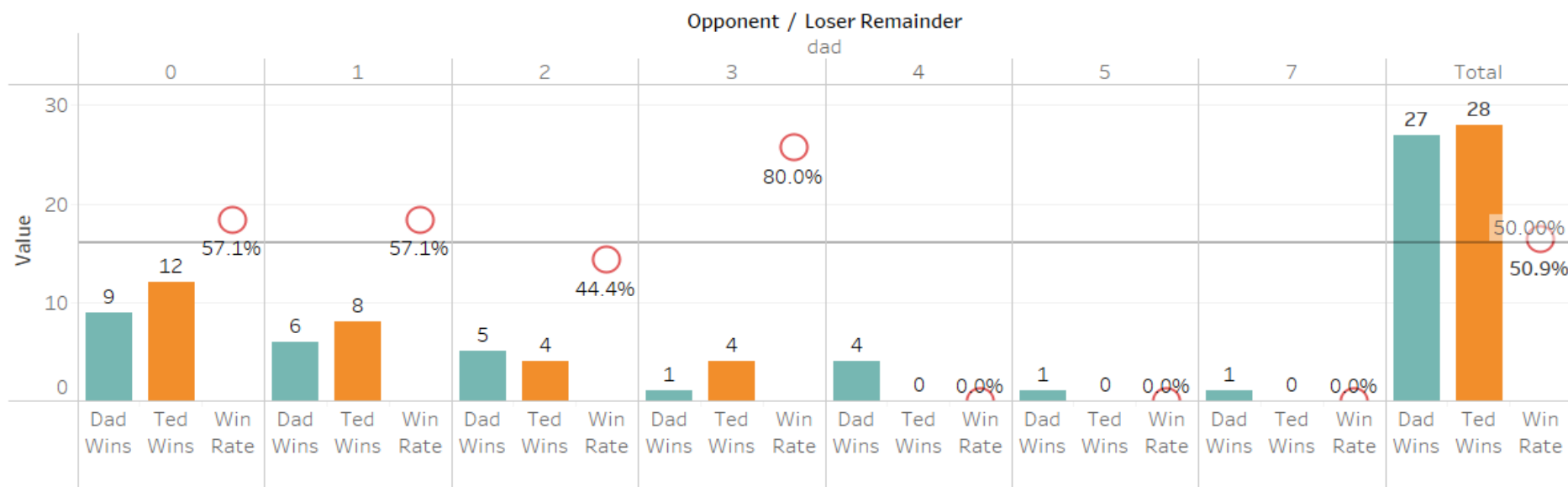


Is Breaking an Advantage?

DadBreak



TedBreak



Location of Games vs Dad

